

Introduction to ethics and the course

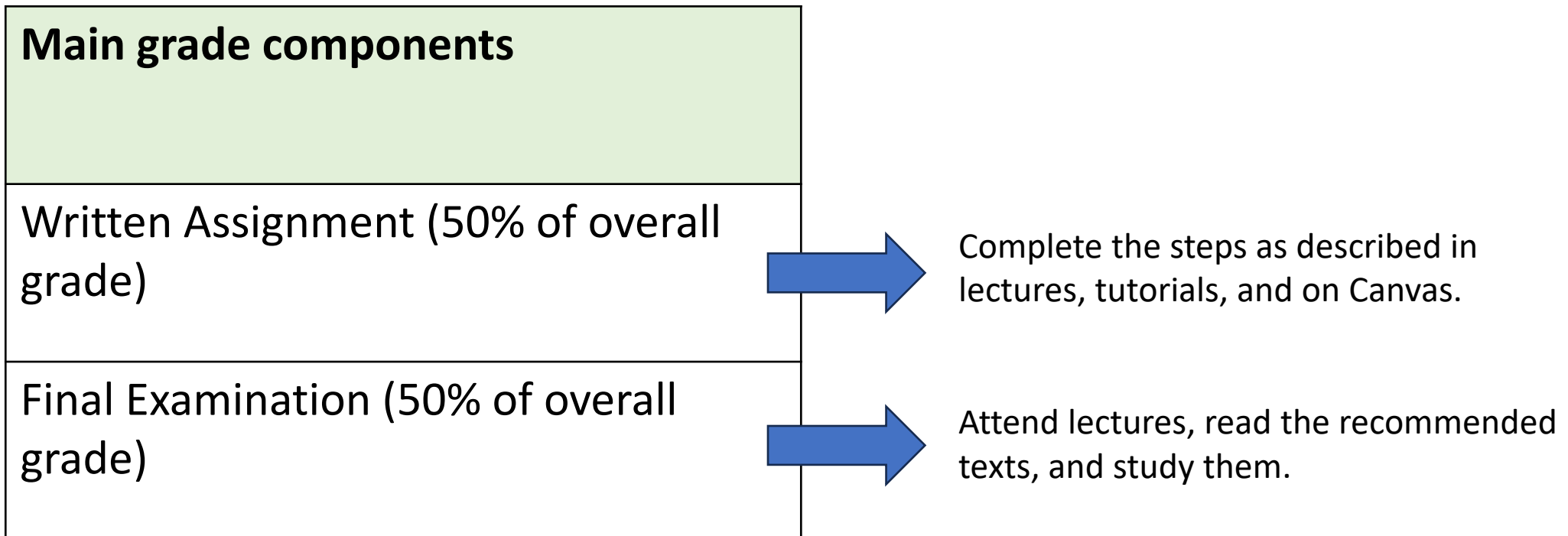
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Introducing the course and topic

Two main topics in the course

- Theory of ethics of AI: lectures and assignment
- Practice of ethics of AI: assignment



Lectures and guided self-study

- In the first 5-6 weeks, there will be two lectures per week covering basic concepts applied to AI. The labels in your TU/e calendar are not very useful. Please look at the schedule on Canvas instead.
- In the first 2 weeks, you will form a group and determine a topic for the assignment.
- Your group will meet with coaches three times during the quarter. You sign up for time slots as a group.
- In the last couple of weeks, you will work on the assignment and study for the exam.
- You should look at the modules each week. These are in the Modules section of the Canvas page.

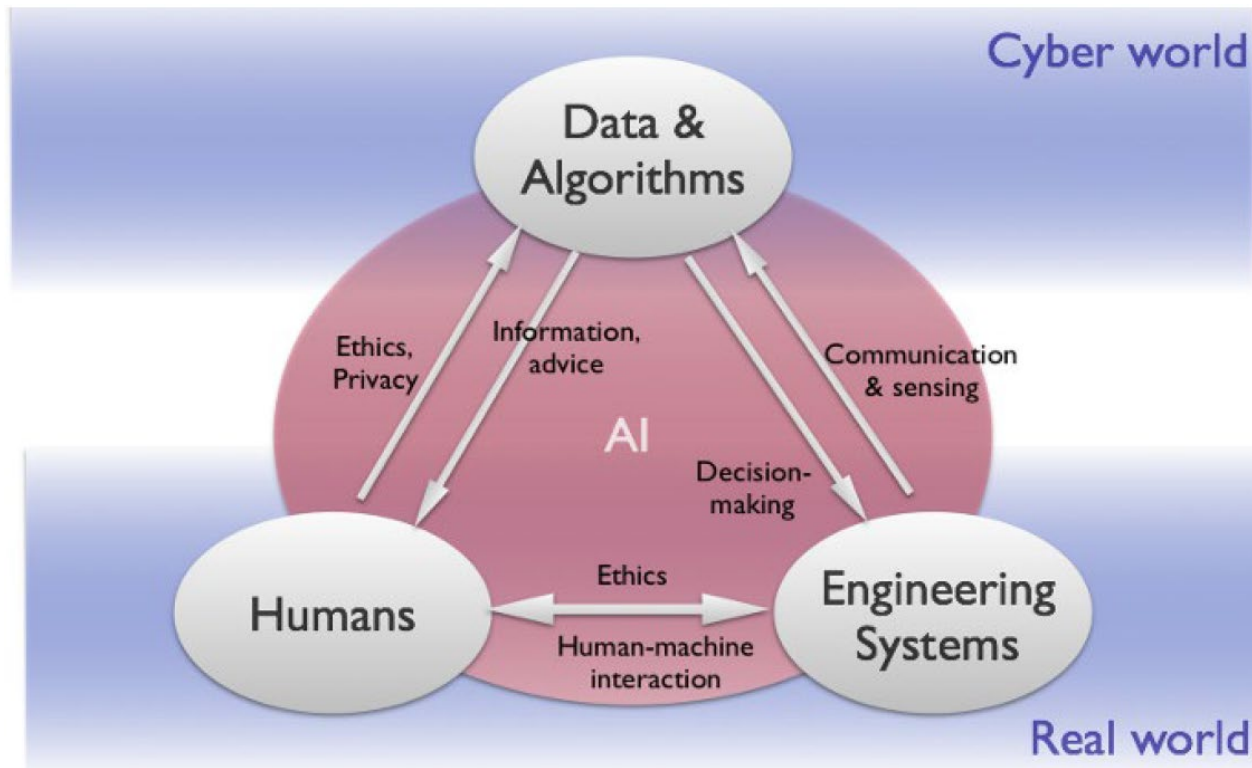
Why study ethics of AI?

- AI systems are meant to support or replace human decision-making. Ethics aims to answer the question how decisions should be made. Hence ethics can be used to guide AI design.
- Some AI designs and features have an ethical purpose (e.g., protecting privacy, achieving fair outcomes). Therefore, studying ethics will help understand and guide these designs and features.
- AI has features of speed, scale, opacity, and independence of human oversight that require special ethical attention.
- There is serious societal controversy and anxiety about AI.

How to study

- Attend the lectures and do the readings.
- Read the material twice, once for orientation, and once after the lectures to identify places where the lectures and readings overlap. The overlap forms the main source of exam questions.
- Exam questions are both open (short paragraph) and closed (MC/ T-F)
- On the exam, there are usually two ways to arrive at a correct answer. The first is to remember points directly from readings and lectures; the second is to be able to reason to the correct answer from your understanding of the material.

Relation to AI&ES program tracks



1. High-tech systems and robotics
2. Mobility
3. Healthcare
4. Smart cities
5. Science and discovery
6. Manufacturing systems

What is ethics?

- Ethics vs. morality
- Ethics is defined as the systematic study of morality. One of its tasks is to establish consistent and well-justified standards for moral judgements and reactions (e.g., assigning praise and blame).
- Law vs. ethics

Structure of ethics

We seek “reflective equilibrium” between the elements here. “Wide reflective equilibrium” refers to the need to take science and empirical factors into account, as well.

- Judgments in concrete cases (morality)
 - E.g., the “toeslagenaffaire” is blameworthy



- Principles, values, virtues (ethics = systematized morality)
 - E.g., Rawls’ Difference Principle; Principle of Utility; Value of Explicability



- Theories (views about which principles, values, virtues have priority, and what justifies them)
 - Rawls’ Theory of Justice; the Capability Approach; Utilitarianism

The written assignment

Assignment Phase 1 (5%) (tue.nl)

Due 18 Sept

5ARFO Ethics of AI: theory and practice > Assignments

> Assignment Phase 1 (5%) 6d View

Assignment Phase 1 (5%) Published Assign to Edit ⋮

You will choose one of the three topics in this [set of case descriptions](#) ↓.

[ANWE: AI for Animal Welfare](#)

[EADS: AI for Eating Disorder Symptoms](#)

[ENSU: AI for Environmental Sustainability](#)

At the end of the first three weeks, you will produce an initial report describing the motivation and goals of the project, identifying requirements, and anticipating issues that need to be addressed. See the suggested format document and assessment criteria (added here soon).

Points 10

Submitting a file upload

Assigned texts/ recommended readings

The primary reading is the first section of the book:

Smuha NA, ed. (2025). *The Cambridge Handbook of the Law, Ethics and Policy of Artificial Intelligence*. Cambridge Law Handbooks. Cambridge University Press. <https://www.cambridge.org/core/books/cambridge-handbook-of-the-law-ethics-and-policy-of-artificial-intelligence/contents/B58479AF2A9C398BD461575A18F828D3>

The exact schedule and readings can be accessed from the home page on Canvas.

Read with the following questions in mind:

What is/ are the main question(s) the authors are addressing?

What are the authors' main conclusions or main claims?

What specific concepts or frameworks do the authors rely on?

What references do they give for those concepts or frameworks?

What evidence or reasons do the authors supply to support the conclusions?

Do you agree with the authors' conclusions?

How might someone object to the authors' arguments?

A photograph of a modern university campus with large glass buildings. In the foreground, a man and a woman are sitting at a brightly colored picnic table (yellow, orange, red, and blue) on a green lawn. They are smiling and talking. In the background, other students are sitting at similar tables, and there are trees and a clear blue sky.

Wellbeing and diversity @ EE

INTRODUCTION SLIDES FOR COURSES

TU/e Vision on Student Wellbeing

TU/e aims to ensure that students are aware of what wellbeing means and feel empowered and enabled to take care of their wellbeing. Wellbeing is the responsibility of our community as a whole. Together we create a welcoming, safe and inclusive environment and provide the necessary knowledge and support to enable students to take control of their own wellbeing.



*Read the full
TU/e vision here*

5 dimensions of student wellbeing



Social



Emotional



Academic



Financial



Physical

“The Electrical Engineering department at TU/e is committed to building an inclusive and welcoming community that values diversity and promotes equity.

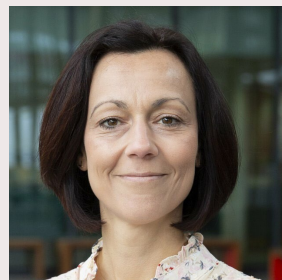
We believe that a diverse and respectful environment enhances innovation, learning, and collaboration. Our goal is to ensure that all students, faculty, and staff feel supported, valued, and empowered to succeed.”

- Prof. dr. ir. Mark Bentum, Dean of Electrical Engineering

Promotors of wellbeing: whom to contact



Overview of all contact persons



Your academic
advisors at EE



THOR wellbeing support page

Fundamental sources: The UDHR and the Belmont Report

Fundamental sources

The fundamental sources I will discuss are systematic ethical standards that have been agreed upon by many human societies:

- *The Universal Declaration of Human Rights* (1948)
- *The Belmont Report* (1979)

These two sources do not mention artificial intelligence at all. But we can still apply them to situations in which AI is being developed or deployed.

The Universal Declaration of Human Rights (UDHR)

- Created as one of the first and most significant acts of the United Nations in 1948.
- Responds to the moral atrocities committed in World War II.
- Written collectively by delegates to the UN. The guiding force behind the process was former US first lady Eleanor Roosevelt.
- Articulates 30 rights of many different sorts.
- Ratified by many countries around the world and has a role in international law. The rights that it proclaims are therefore both moral and legal.
- Human rights are “rights possessed by all human beings (at all times and all places), simply in virtue of their humanity” (Simmons 2001).

Example

Article 12

No one shall be subjected to arbitrary interference with his privacy, family, home or correspondence, nor to attacks upon his honour and reputation. Everyone has the right to the protection of the law against such interference or attacks.

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- Articles 7 & 8 of the EU Charter of Fundamental Rights (2000)

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Case law informed by
theory

Implementation and
enforcement

Logical relationships between rights and duties

General idea: Negative or positive right $\leftrightarrow$ Negative or positive duty

R has a right to X $\rightarrow$ S has a duty not to interfere with X (negative right/ duty)

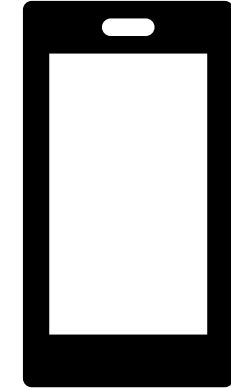
R has a right to X $\rightarrow$ S has a duty to provide X (positive right/ duty)

Examples: right to have a family, right to vote, right to political assembly

Note that S can refer to different or multiple entities:

usually fellow citizens, institutional entities, and/or the government.

Do others have a duty not to interfere with R having X? Are violations punishable?



How strong is R's right to X?

Do others have a duty to assist R X-ing?

Does the right to X override other rights?



Does society have a duty to help pay for R's X-ing?

Or its consequences?

S. Matthew Liao, Human rights and AI

Fundamental Conditions Approach [FCA]: “human rights protect the fundamental conditions for pursuing a good life” (2020, 15).

Human rights have serious implications for the use of AI:

- “companies and ... AI researchers are responsible for ensuring that they do not use the user’s data in ways that could undermine that user or some other users’ human rights” (ibid., 17).
- At the societal level, solutions may be needed: “one proposed solution to automation [and resulting job loss] is a universal basic income...” (ibid., 18).

Consider AI risks related to the UDHR human rights

- Read and think about some of the human rights in the UDHR. You can find a link at the Modules page, or simply search.
- In your opinion, what are the human rights most likely to be *supported* by the development of AI in the next 30 years? What are the human rights most likely to be *threatened* by the development of AI in the next 30 years?
- Please describe a short example of an AI application that would support or threaten one of the UDHR rights.

Second Fundamental Source: The Belmont Report

- [The Belmont Report | HHS.gov](https://www.hhs.gov/ohrt/belmont-report)

THE BELMONT REPORT

Office of the Secretary

Ethical Principles and Guidelines for the Protection of Human
Subjects of Research

The National Commission for the Protection of Human Subjects of
Biomedical and Behavioral Research

April 18, 1979

AGENCY: Department of Health, Education, and Welfare.

ACTION: Notice of Report for Public Comment.

Differences and similarities with the UDHR

Unlike the UDHR,

- The Belmont Report has no official status in itself; it is influential because it has been adopted as the general ethical framework for human research ethics in the USA and Europe.
- The Belmont Report is framed in terms of ethics principles rather than rights.

Like the UDHR,

- The Belmont Report is multiple-authored and results from a process of consensus building.
- The Belmont Report was written as a response to serious moral failures.

“Four Principles” Approach

Beneficence

Non-
maleficence

Respect

Justice

Emergence
of human
research
ethics in the
USA



<http://www.tuskegee.edu/sites/www/Uploads/images/TopNews/Syphllis-survivors-with-Clinton1.jpg>



Rothman, D.J. Were Tuskegee and Willowbrook 'studies in nature?'
Hastings Center Report 12: 1982

What makes Tuskegee and related cases wrong?

The most obvious question, which has not received sufficient attention, is: how could an experiment that seems so clearly to violate medical ethics have been started and have continued for so long? It is both too simple and condescending to the researchers to argue that the primitive state of medical research ethics in the 1930s excuses the venture. As early as 1865 Claude Bernard asserted this familiar and still valid dictum: "The principle of medical and surgical morality, therefore, consists in never performing on man an experiment which might be harmful to him to any extent, even though the result might be highly advantageous to science, i.e., to the health of others."

Respect and Justice Principles were innovations meant to deal with such cases

Intuitively, there was something wrong with Tuskegee from its inception: “using people”.

The principles of beneficence and non-maleficence do not fully explain this wrong.

Hence, we need one or more additional principles to explain this wrong.

[Nuremberg Code - UNC Research](#)

Human subjects research today

In order to prevent scientific misconduct as well as to ensure that scientific research promotes the “greater good”, all human subjects research must be approved by an **Ethical Review Board** (ERB).

- [TU/e Ethical Review Board](#)

ERB approval is mandatory for all human subjects research, most notably **biomedical** and **psychological** research.

- But what about **AI** research involving humans?

AI as human subjects research

Artificial intelligence conducts **human subjects research** insofar as:

- The data it processes is *about* humans.
- The knowledge being acquired is used to drive decisions and develop technologies that *affect* human beings.
- Moral problems in this field are especially tricky if unequal power relationships are involved.

However, only human subjects research done with government funding, or in certain domains, is legally subject to institutional ethics review.

Facial recognition and respect for persons

“Data without consent

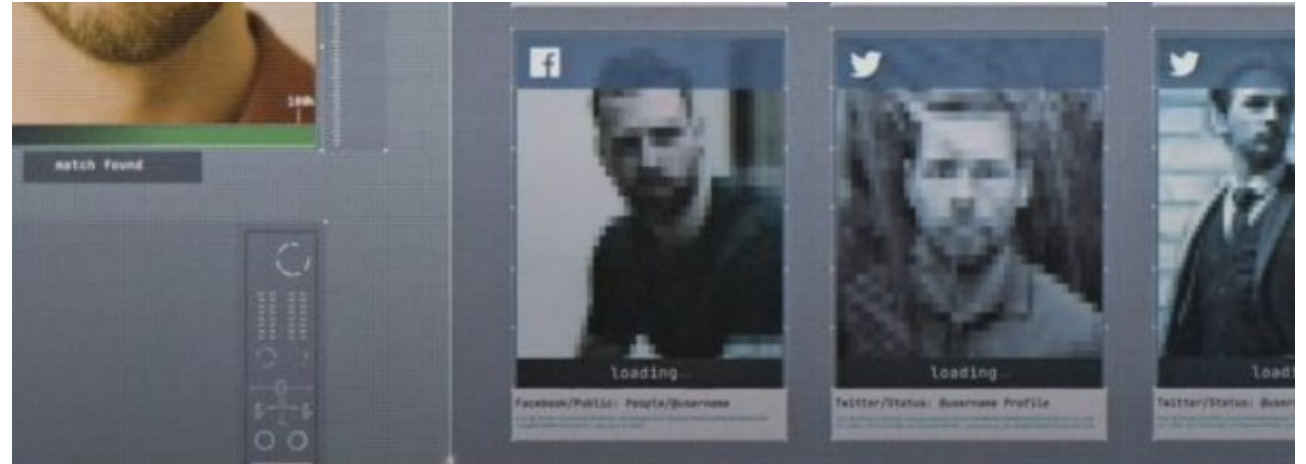
For facial-recognition algorithms to work well, they must be trained and tested on large data sets of images, ideally captured many times under different lighting conditions and at different angles. In the 1990s and 2000s, scientists generally got volunteers to pose for these photos — but most now collect facial images without asking permission.

For instance, in 2015, scientists at Stanford University in California published a set of 12,000 images from a webcam in a San Francisco café that had been live-streamed online². The following year, researchers at Duke University in Durham, North Carolina, released more than 2 million video frames (85 minutes) of footage of students walking on the university campus³.”

Richard van Noorden, “The ethical questions that haunt facial-recognition research”, *Nature* [News Feature], 18 November 2020. <https://www.nature.com/articles/d41586-020-03187-3>

In the news

- [Boete van privacywaakhond voor verzamelaar van miljarden foto's van gezichten \(nos.nl\)](#)



NOS Nieuws • Vandaag, 09:06



Boete van privacywaakhond voor verzamelaar van miljarden foto's van gezichten

Het omstreden Amerikaanse bedrijf Clearview AI, bekend van zijn gezichtsherkenningssoftware, krijgt een boete van 30,5 miljoen euro van de Autoriteit Persoonsgegevens (AP). De privacywaakhond zegt dat Clearview een illegale database heeft opgebouwd met miljarden foto's van gezichten, ook van Nederlanders.

What can we infer from the Four Principles about facial recognition research using public data?

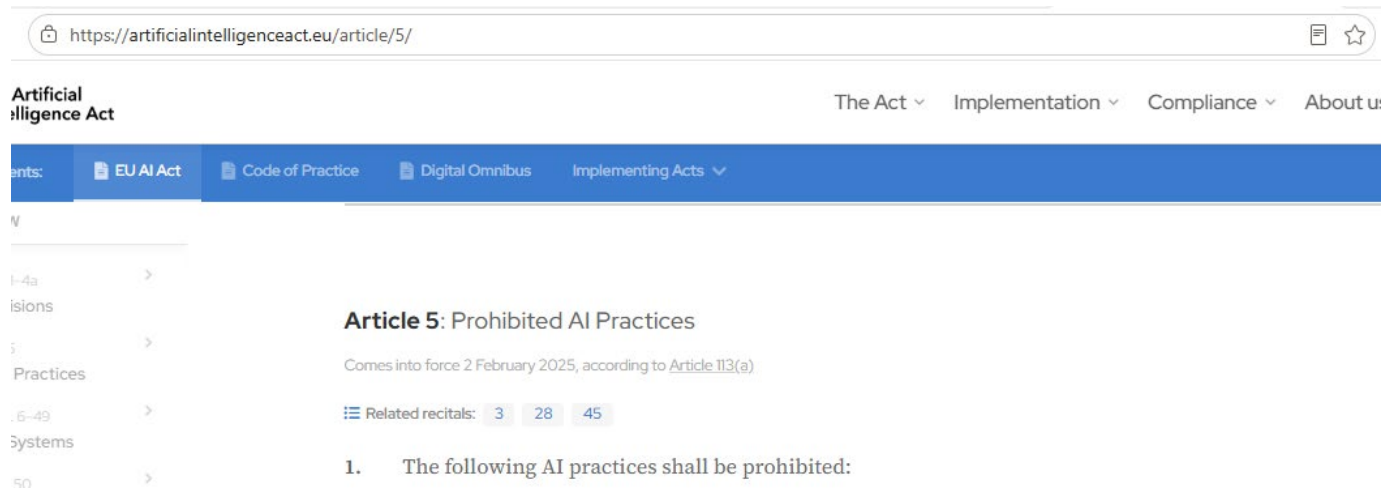
- Non-maleficence
- Beneficence
- Respect
- Justice

“Respect for persons incorporates at least two ethical convictions: first, that individuals should be treated as autonomous agents, and second, that persons with diminished autonomy are entitled to protection. The principle of respect for persons thus divides into two separate moral requirements: the requirement to acknowledge autonomy and the requirement to protect those with diminished autonomy.”

“Who ought to receive the benefits of research and bear its burdens? This is a question of justice, in the sense of "fairness in distribution" or "what is deserved." An injustice occurs when some benefit to which a person is entitled is denied without good reason or when some burden is imposed unduly.”

[Belmont Report]

Facial recognition in the AI Act



- (e) the placing on the market, the putting into service for this specific purpose, or the use of AI systems that create or expand facial recognition databases through the untargeted scraping of facial images from the internet or CCTV footage;

The EU Approach to AI Policy

- The following elements are part of the general framework for the EU's approach to AI policy:
 - General approach: RISK-Based (See link to AI Act summary below)
 - Rights-based. The restrictions on prohibited and high-risk AI systems are justified by way of individual (human) rights.
 - Ethical AI:
 - Based on “Four Principles Plus One” but expanded to include human oversight, safety, privacy, transparency, fairness, and sustainability (see CHLEP Ch. 3, p. 63).
 - The “Plus One” is *explicability* (explainability).

<https://artificialintelligenceact.eu/high-level-summary/>

What is *not* included (much) in the UDHR and the Belmont Report?

- Environmental ethics/ sustainability values considered as such
- Animal ethics
- Ethics of non-human intelligent beings
- Ethics of risk
- Virtues and values insofar as these go beyond rights and rules

Questions?



Assignment topics

- You can sign up in person for a topic on Thursday 3 Sept.
- Work is done in groups of four.
- There is a limited number of slots per topic. You might not be able to work on your most preferred topic.
- There are client meeting on Monday afternoon/ early evening. These are meant to help you get ideas about what tools might be most useful, fun, or promising.

Topic 1: **ANWE**: AI for Animal Welfare

- In this assignment topic, you will develop a design for a tool that **manages animal welfare**.
- You will need to select a destination context such as agriculture, wild animal management, or laboratory research on animals.

Client Q&A: Monday at 15:00 online

Topic 2: EADS: AI for Eating Disorder Symptoms

- Clinically tested interactive tool
- Currently the chat interface is not AI-based. The question is how an AI tool could be developed to support patients, family members, and/or the public on this topic
- Client Q&A on Monday at 17:30

fEATback

Voor iedereen die zich zorgen maakt
over eten of gewicht

Weet jij alles over eten?

Sarah, Kai en Robin eten alledrie
anders.

Kun jij per situatie voor hen
**beoordelen of ze die goed
aanpakken?**



Topic 3:

ENSU: AI for Environmental Sustainability

In this challenge topic, students will draft a new Sustainability Tool for TU/e concerning Research, Education or Operation. It can be based on the existing sustainability self-assessment tool, but it should incorporate AI for sustainability in a creative, well-supported way. It should also be consistent with TU/e sustainability policy.

* Client Q&A on Monday at 18:30

First steps of the assignment

- Gradually form a group of four, preferably a group with multidisciplinary perspectives and multiple academic and cultural backgrounds. You can't finalize this today, so stay open about it.
- Learn about the ***target context***. What are the problems and challenges that practitioners and other stakeholders have in that context?
- Do background research on academic and grey literature regarding these problems.
- Avoid “techno-solutionism”: “the problem definition follows the solution and is twisted to fit the solution, rather than the other way around” (Siffels & Sharon 2024, p. 3)

Orientation activity

- See the canvas survey, linkable from Modules

Please introduce yourself to at least four other people during the self-study period on 1 September, and answer the survey questions.

Please complete the survey.

- For today, if you've met enough people and gotten their details, you are free to finish up whenever you are done.